

SymPy Bug Card

Bug 51: False solution for $\text{atan2}(x, 1) = \pi$ in `solveset`

Task. Solve the real equation

$$\text{atan2}(x, 1) = \pi.$$

SymPy's answer.

$$\text{solveset}(\text{atan2}(x, 1) = \pi, x, \mathbb{R}) = \{0\}.$$

Correct answer.

$$\emptyset.$$

Explanation. Substituting the returned value gives $\text{atan2}(0, 1) = 0$, not π . Since the second argument is positive, $\text{atan2}(x, 1) = \arctan(x)$, whose real principal range is $(-\pi/2, \pi/2)$.

Likely root cause. The diagnosis traces the false positive to SymPy's generic `_invert_real` path in `solveset`: after $\text{atan2}(x, 1)$ simplifies to $\arctan(x)$, it applies $\tan$ to the out-of-range right-hand side π without enforcing $\arctan$'s range. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-051-false-solution-for-atan2-x-1-pi-in-solveset/](#)